PHASE 02 / ARCHITECTURE AND DATA FLOW

Architecture and data flow

Keep the browser, API, recommendation provider, and persistence responsibilities small and explicit.

Activities

Serve the responsive HTML, CSS, and JavaScript from FastAPI so the browser and API share one origin.

Route planner requests to separate home, party, and jewelry inputs; keep shared recommendation orchestration in a service module.

Use typed Pydantic models to validate requests and response structures at the API boundary.

Use SQLite for accounts and saved recommendations; keep image bytes transient and do not store outfit photos in history.

Use one Gemini service client when configured and a local fallback provider otherwise; build platform search URLs without scraping stores.

Phase outputs

Acceptance checks

Each component has one main responsibility and an agreed input/output.

The browser calls same-origin `/api` routes without exposing provider secrets.

Provider output is normalized before it reaches the interface.